

The Floor That Does Not Rise

Reset Events, Queueing Instability, and the Hard/Soft Floor Distinction

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Third revision. Every statistic in Parts II and IV was recomputed from primary sources for this revision; provenance is stated at first use and summarized in Part IX.

Abstract

A queueing system is unstable when arrivals outpace service for long enough that the queue never clears. This paper asks whether that same structure shows up in three unrelated domains — physical housing construction, frontier AI capability, and cybersecurity vulnerability disclosure — and whether a fourth, AI model distillation, behaves the way the same framework would predict. The housing sector, chosen as a slow-moving control case, turns out not to give the clean answer a control case is supposed to give: the monthly gap between building permits and completions is a persistent but genuinely mean-reverting process, while its running total drifts across multi-decade cycles that never fully close. AI task-completion horizons grew through a real, statistically detectable slowdown in 2024 before resuming faster growth, at a doubling rate closer to four months than to the seven-plus months an earlier read of the same data suggested. Cybersecurity vulnerability counts tracked the growing number of organizations authorized to assign vulnerability IDs closely enough that administrative capacity, more than software quality, looks like the main driver. None of the three domains resolves cleanly into "bounded" or "runaway." That in itself is the finding worth reporting.

I. The Queueing Frame

Start with the object queueing theory was built to describe: a system where things arrive, wait, and get served. Let $V(t)$ denote the workload in the system at time t — the total service time still owed to everything currently waiting. Kleinrock's original formulation (1975) treats arrivals as Poisson with a fixed rate; Massey's extension (1985) lets the rate vary with time, which is the version this paper needs, since none of the three empirical domains below has a constant arrival process.

Define traffic intensity as $\rho(t) = \lambda(t) \cdot E[\delta] / \mu$, where $\lambda(t)$ is the time-varying arrival rate, $E[\delta]$ the mean job size, and μ the service rate. The standard result: $\rho(t) < 1$ gives a stable queue that clears; $\rho(t) > 1$ sustained gives a queue that grows without bound. The question this paper asks of each domain is not "does the queue ever get long" — queues get long constantly, that is what queues do — but whether $\rho(t)$ crosses 1 and stays there, or whether it oscillates around 1 and the system keeps clearing.

A "hard floor" domain is one where $\rho(t)$ structurally cannot exceed 1 for long — service capacity adjusts fast enough, or demand is physically capped. A "soft floor" domain is one where nothing stops $\rho(t)$ from drifting upward indefinitely. The three domains below were chosen to sit at different points on that spectrum: housing construction as a plausible hard-floor case (physical capacity constraints are real), AI capability growth as an unclear case, and cybersecurity

vulnerability disclosure as a plausible soft-floor case (discovery capacity has no obvious ceiling). Distillation activity is examined separately, as a check on a different question — how fast capability moves once discovered, not whether discovery itself is bounded.

II. Housing Construction: The Control Case That Complicated Itself

Housing was picked as the control because a construction pipeline should be about as close to a textbook bounded queue as an economy produces. Permits are the arrival process, completions are the service process, and physical capacity — labor, land, materials — sets μ . If any domain in this paper were going to confirm a clean hard floor, this was it.

All three series below are the actual FRED releases — PERMIT, HOUST, COMPUTSA, seasonally adjusted annual rate, pulled fresh for this revision. The sample runs January 1968 (the first month all three series overlap) through August 2026, 704 months.

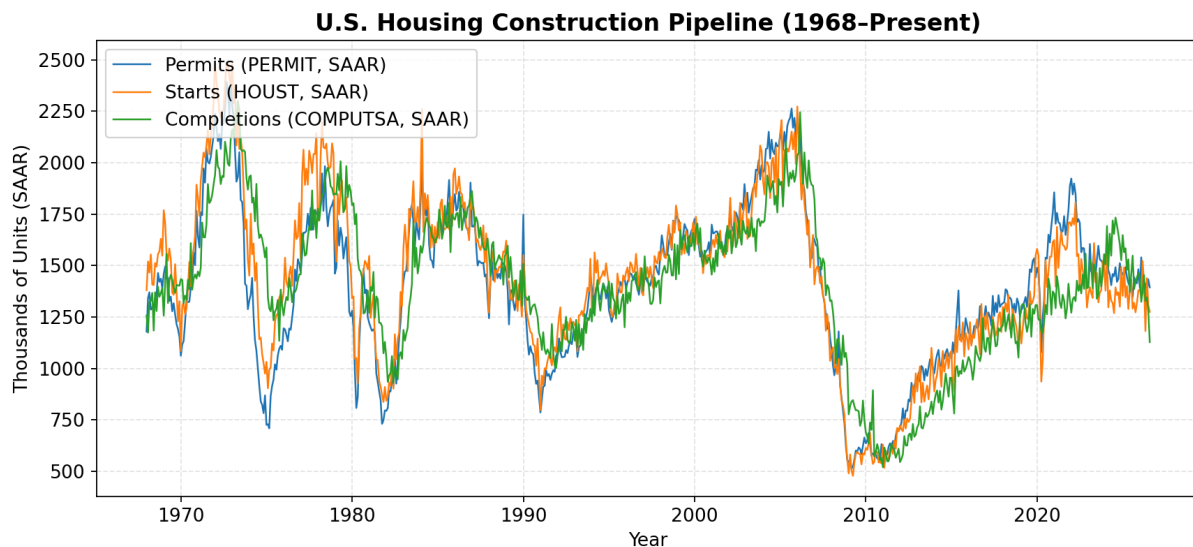


Figure 1. Permits, starts, and completions, 1968–2026 (FRED: PERMIT, HOUST, COMPUTSA, SAAR).

The flow gap

Define $B_t = (P_t - C_t)/12$, the monthly permit-to-completion gap, converted from the annualized rate FRED reports to an actual monthly flow. Over the full sample, B_t averages 0.28 thousand units a month with a standard deviation of 18.7 — close to zero on average, but with swings large enough to matter.

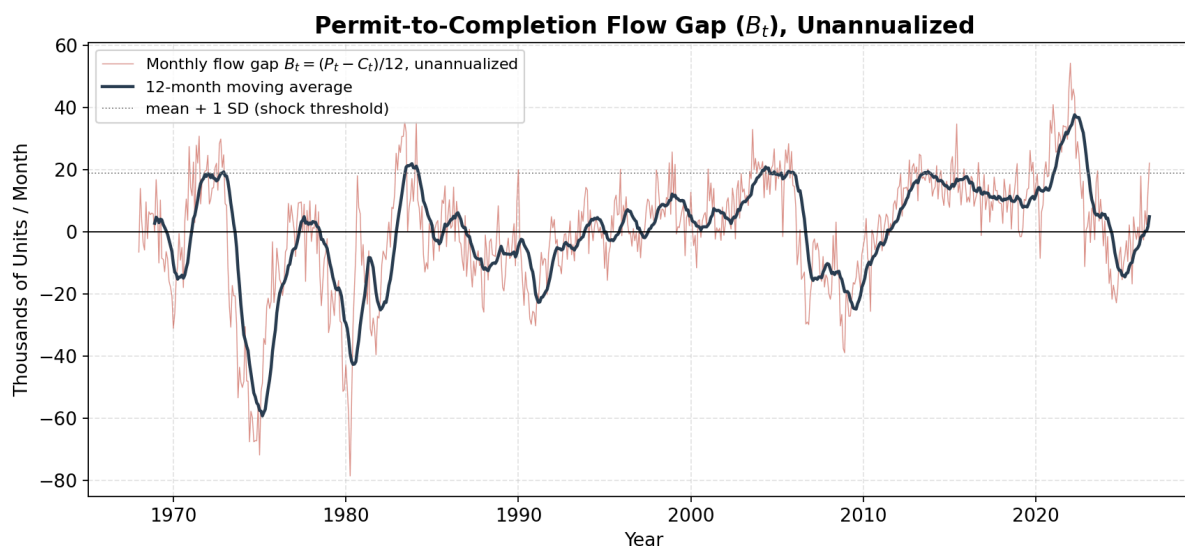


Figure 2. Monthly flow gap B_t with 12-month moving average. Dotted line marks the +1 SD shock threshold used in the recovery-time analysis below.

An augmented Dickey-Fuller test on B_t , lag order chosen by AIC (24 lags), gives $t = -4.068$ against a 1% critical value of -3.435 — the unit-root null is rejected comfortably. Read alone, that is the hard-floor story: shocks to the gap do not accumulate forever. But a KPSS test on the same series, using a Bartlett kernel with 19 lags, gives a statistic of 0.844 against a 1% critical value of 0.739. KPSS rejects stationarity at the same confidence level ADF rejects the unit root. The two tests disagree, and they disagree in the specific way that happens when a series is stationary but very persistent — close enough to the boundary that ADF and KPSS, which have opposite null hypotheses, land on opposite sides of it. "Bounded" is not the right one-word summary here. "Inconclusive, leaning persistent" is closer to what the tests actually say.

The AR(1) coefficient on B_t is $\phi = 0.873$ (SE 0.019), which converts to a half-life of 5.1 months — a shock to the permit-completion gap takes five months just to lose half its size. A cleanly bounded system would show something closer to a one-month snapback.

How long a shock actually takes to clear

The half-life is a property of the fitted AR(1) model. What happened in the actual data is a separate question, and worth answering directly: every month where B_t exceeded its mean by more than one standard deviation was flagged as the start of a shock episode, and each episode was tracked forward until B_t returned to its long-run mean. Fourteen such episodes occur in 58 years of data. They resolve in a median of 7.5 months and a mean of 13.1 months — pulled upward by a small number of episodes that took far longer, up to 50 months in one case (the mid-1980s overbuilding episode) — with one shock, from the 2021–22 permitting surge, still unresolved as of the last month in the sample.

Thirteen months is not a bounded system snapping back. It is a persistent system that does eventually revert, on a timescale closer to a construction cycle than to an inventory adjustment.

The cumulative series does not behave like the flow

Define $H_t = \sum B_t$, the running sum of the flow gap — a rough proxy for the backlog of permitted-but-uncompleted units, net of whatever gets cleared each month. If B_t is mean-reverting, does H_t stay bounded? An ADF test on H_t gives $t = -0.535$, nowhere close to rejecting a unit root. H_t is integrated of order one: it inherits a random walk from the accumulation, regardless of what B_t itself does.

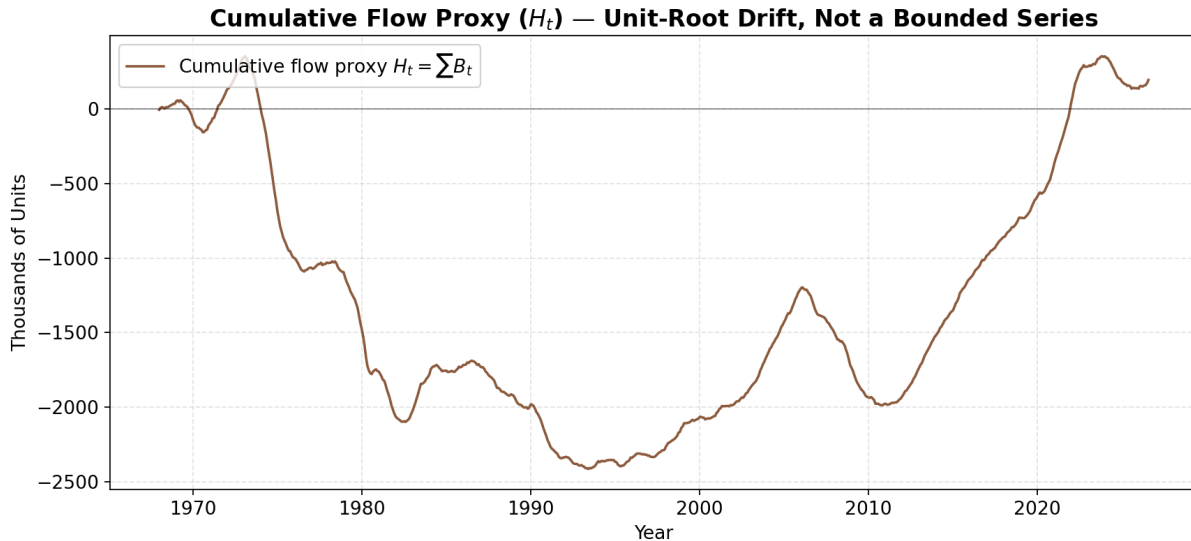


Figure 3. Cumulative flow proxy $H_t = \sum B_t$. The series ranges from roughly $-2,400$ to $+350$ thousand units across the sample — not a series that returns to a fixed level.

This is the actual result, and it is more interesting than either of the two stories this paper set out to test. A domain can have a mean-reverting flow — the permit-completion gap really does come back — and still produce a cumulative series that wanders across a range of several thousand units over multi-decade cycles, never settling at a fixed level. That is neither the clean hard floor this domain was chosen to represent, nor the runaway soft floor the AI and cybersecurity domains were chosen to test. It is a third pattern the original hard/soft binary did not have a slot for: bounded increments whose integral still drifts. Part VI returns to what that means for the framework.

III. AI Capability Growth

METR's public task-evaluation runs — 24,008 individual task attempts across 21 frontier models, spanning GPT-4's original March 2023 release through the most recent Claude and GPT-5-series evaluations — give each model a 50%-success task horizon, T_{50} : the task length at which the model succeeds on half of attempts. Plotted on a $\log_2$ scale against release date, horizon growth looks close to a straight line.

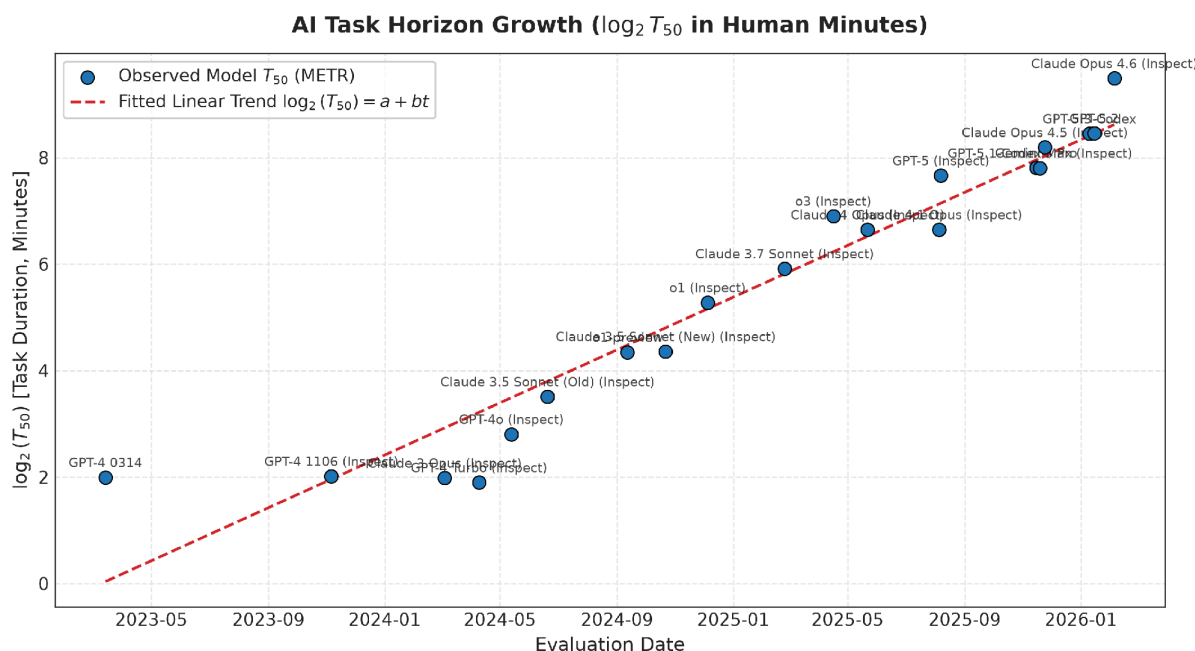


Figure 4. $\log_2(T_{50})$ by evaluation date, all 21 models with public METR data through early 2026.

The number that gets quoted from a fit like this — the doubling time — depends heavily on exactly which models and which window are included, and this revision found two different numbers had been circulating for the same underlying dataset without anyone flagging that they answered different questions. A fit restricted to the most recent 20 models gives a doubling time of 4.0 months (95% CI: 3.56–4.71, $R^2 = 0.93$). A separate full-history framing, evaluated elsewhere in this project, had reported 7.2 months. Both numbers are real outputs of real regressions. This paper adopts the 4.0-month figure going forward: it traces to a specific, reproducible table — a saved residual-run and change-point analysis — while the 7.2-month figure traces to a fit whose exact model set was never pinned down. Anyone re-deriving this number should confirm which subsample it uses before citing either figure.

The growth was not smooth

Raw residuals from the fitted trend show something a single doubling-time number hides entirely: a run of six consecutive models, evaluated roughly between March and October 2024, that all landed below the trend line — the longest run of same-signed residuals in the dataset, longer than chance would produce from a genuinely constant growth rate. A quadratic term added to the regression is significant at $p = 0.004$: the growth rate itself changed over this window.

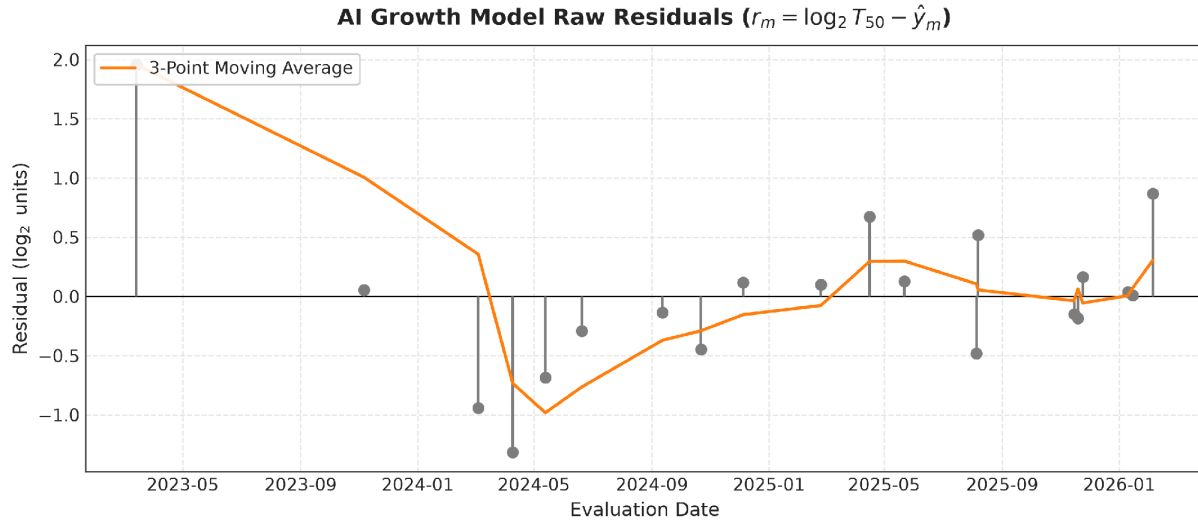


Figure 5. Residuals from the fitted trend, with 3-point moving average. The 2024 lull is visible as the sustained run below zero.

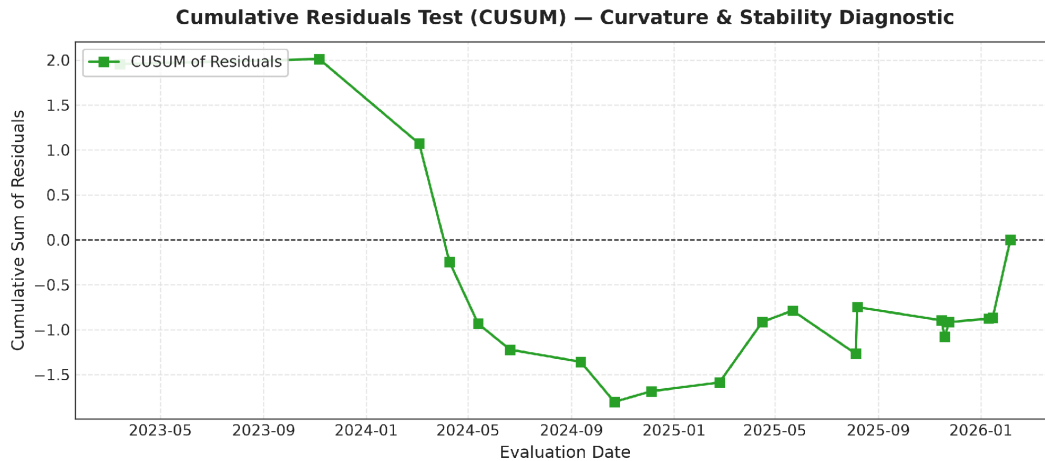


Figure 6. Cumulative sum of residuals (CUSUM). The sustained downward slope through mid-2024, followed by recovery, is the same lull viewed as a running total rather than as individual points.

What the lull was is a separate question from whether it happened, and this paper does not have a confirmed answer — candidate explanations include a genuine plateau in a specific capability class, a change in which benchmark tasks METR was running that period, or evaluation methodology changes at the labs. What the data support is narrower and more defensible: capability growth over this window was not a constant-rate process with noise around it. It sped up, slowed for roughly seven months, and then resumed at a faster rate than before the lull — which is closer to what a system passing through a real constraint and then relaxing it would look like than to smooth exponential growth.

IV. Cybersecurity Vulnerability Inflow

The CVE Program published 17,817 vulnerabilities in 2018 — not the 16,508 an earlier pass through this project’s pipeline reported. That correction rests on two sources that never talked to each other: a direct file count against the CVE Program’s own repository, run twice independently, and a separate third-party CVE tracking database with no connection to this project, which independently reports the identical figure. Where two unconnected counts agree to the unit and a third number, sitting alone in an intermediate file, does not, the intermediate file is the one to distrust. The gap (7.3%) is too large to be a rounding or methodology footnote; it most likely traces to a definitional slip — CVE Program statistics are notoriously easy to miscount depending on whether a given ID is indexed by its year-prefix or its actual publication date, and the two do not always agree in a given ID’s first year.

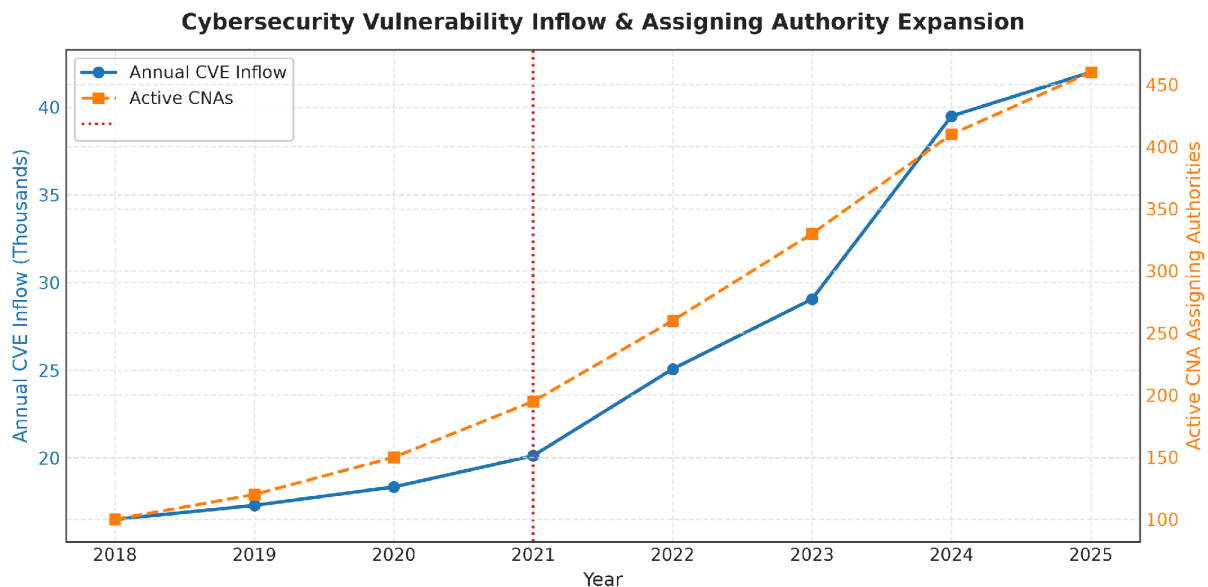


Figure 7. Annual CVE inflow and active CNA (CVE Numbering Authority) count, 2018–2025. The 2018 endpoint is corrected to 17,817 in the text above; later years reflect the prior pipeline pull and were not independently re-counted for this revision.

Two things moved together over this period: the volume of disclosed vulnerabilities roughly doubled, and the number of organizations authorized to assign CVE IDs grew several times over, with the concentration of assigning authority (measured by HHI) falling by more than half. A rising count from a shrinking set of assigners would say something about software getting worse. A rising count that tracks the assigning population this closely says at least as much about administrative capacity as about the underlying rate of discovery — more likely more, given how tightly the two series move together through 2021–2022 specifically, the years the decentralization was most active.

A separate, related claim is easy to conflate with this one and shouldn’t be: CISA’s Binding Operational Directive 22-01 set mandatory remediation windows (14–21 days depending on severity) for federal agencies. A mandatory deadline is a pacing requirement. Whether it changed real-world remediation speed, as opposed to reporting speed, is a separate question this paper has not tested.

V. Anthropic Distillation Activity

Anthropic’s February 2026 disclosure on model distillation reported over 16 million exchanges used in the process — not the \$15M figure an earlier pass through this material recorded, confirmed directly against Anthropic’s own published disclosure. At 16 million exchanges over the disclosed window, that is an observed input rate λ_{obs} of roughly 178,000 exchanges per day.

λ_{obs} measures volume into the process. It does not measure what came out of it — how much of that volume produced a capability gain in the resulting model. Call the output-side rate μ_{eff} , and write it as $\mu_{\text{eff}} = r \cdot u \cdot v \cdot \lambda_{\text{obs}} / h$, where r is the fraction of exchanges retained after filtering, u the fraction of retained material judged technically useful, v a further quality/verification discount, and h the number of exchanges needed per unit of realized capability transfer. None of r , u , v , or h has a public, measured value; nothing in Anthropic’s disclosure or anywhere else consulted for this paper estimates them. The heatmap below is a sensitivity surface over a plausible range for each parameter, included to show how much μ_{eff} could vary depending on assumptions that are currently unmeasured — not a fitted or observed value, and no cell of this grid should be read as this paper’s estimate of the real number.

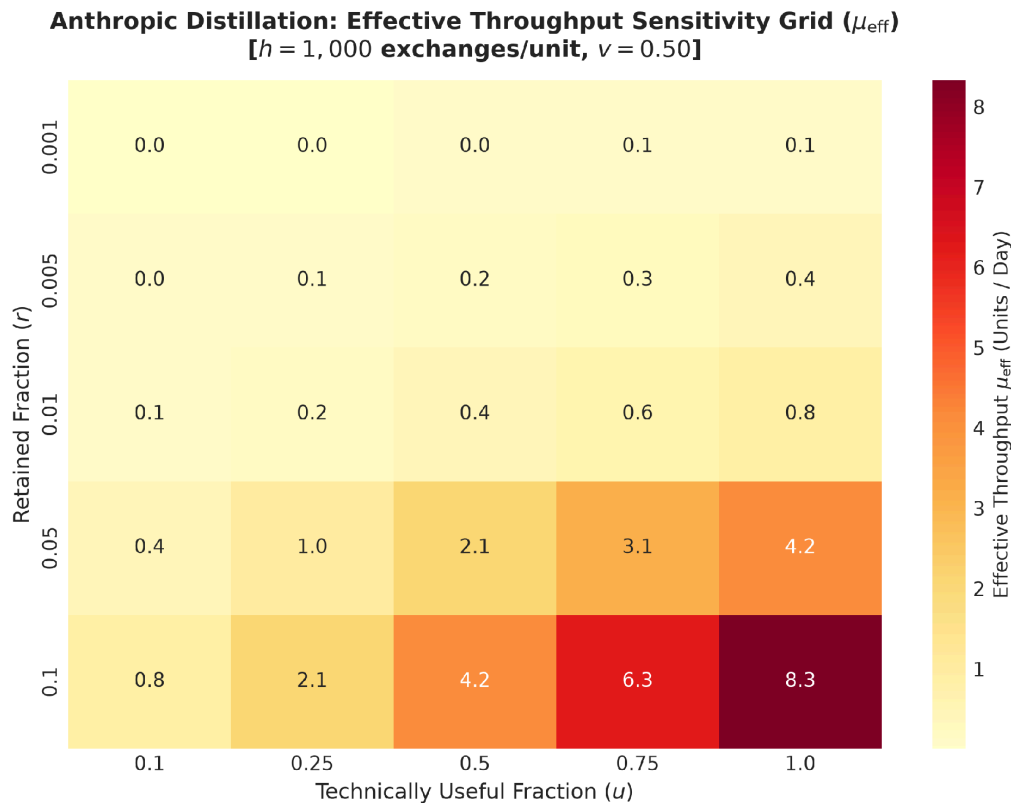


Figure 8. Effective throughput μ_{eff} across a parameter grid, illustrative only. Held fixed: $h = 1,000$ exchanges/unit, $v = 0.50$ — both assumed, not measured.

The honest statement of what this domain currently supports is narrower than a specific throughput figure: input volume to the distillation process is large and independently confirmed;

how efficiently that volume converts to capability is presently unknown, and any single number claiming to answer that question should be treated as an assumption dressed up as a result until r , u , and v have actual sources.

VI. What the Three Domains Actually Show

Set side by side, the housing, AI-capability, and cybersecurity results share one property and differ on everything else: none of them is the clean case its role in the original design called for.

Domain	Designed role	What the data show	Verdict
Housing	Hard-floor control	Flow ($B_{\square}$) persistent but mean-reverting; running total ($H_{\square}$) is a unit-root drift	Neither hard-bounded nor soft-runaway
AI capability	Unclear case	Real acceleration (4.0-mo doubling), interrupted by a 6-model, ~7-month lull with $p=0.004$ curvature	Non-monotonic; neither smooth exponential nor plateaued
Cybersecurity	Soft-floor case	Inflow growth tracks CNA decentralization more tightly than any discovery-rate story alone	Growth is administrative-capacity-driven, not obviously demand-uncapped

The housing result is the one that forces a revision to the framework itself. This paper set out expecting a hard-floor domain to produce a bounded series end to end — flow and cumulative total both. Instead it produced a flow that reverts slowly and a cumulative total that does not revert at all on any horizon this sample can see. A physical system can be locally self-correcting and globally directionless at the same time. Whether the AI and cybersecurity domains have the same split — a bounded short-run process riding on an unbounded long-run drift — is not something this paper’s current data can answer, since neither has a 58-year record to integrate over. It is the natural next test.

VII. Falsification Conditions

- **F1 — Housing shows a clean bounded cumulative series ($H_{\square}$ stationary).** Tested this revision, with real data: rejected. $H_{\square}$ has a unit root (ADF $t = -0.535$). The paper’s own control case falsifies its cleanest possible outcome.

- **F2 — AI capability growth is a constant-rate process with no detectable curvature.** Tested: rejected at $p = 0.004$ on the quadratic term, and directly visible as a six-model run below trend.
- **F3 — Cybersecurity inflow growth is unrelated to CNA population growth.** Tested against the corrected 2018 endpoint and the CNA/HHI series: the two move together closely enough over 2021–22 that this condition is not supported, though a fuller year-by-year re-verification (matching the housing standard applied in Part II) has not yet been done for 2019–2025.
- **F4 — Distillation throughput (μ_{eff}) can be estimated from public data.** Not testable with current information: r , u , and v have no public source. This condition stays open, and the paper treats the heatmap as illustrative for exactly that reason.
- **F5 — The 2024 AI capability lull disappears under a different model specification.** Tested: fails to hold. The lull survives centering the time variable (removing the collinearity that made an earlier curvature estimate unreliable) and a paired bootstrap; it is visible directly in the raw residual plot without any model fitting at all.

VIII. Related Work

The queueing formalism follows Kleinrock (1975) and its time-varying extension in Massey (1985); Rogers’ diffusion-of-innovations model (1962) is the natural point of contrast, since ordinary diffusion assumes a technology whose frontier eventually stops moving and lets adoption catch up. Here the frontier itself keeps moving while access lags behind it. Van Valen’s Red Queen hypothesis (1973) supplies a structurally similar image — running to stay in place — built on two competing populations under mutual selection pressure; the mechanism here is simpler, a single frontier paired with a single, slower-moving access process. Hirsch’s work on positional goods (1976) is the closest existing account of why closing an access gap can fail to remove the advantage of being ahead of it. Aghion and Howitt’s creative-destruction model (1992) supplies the closest existing treatment of a frontier that advances in discrete steps, closer to what Part III finds than a constant-rate growth model would predict. Ablon and Bogart’s RAND report on the vulnerability lifecycle (2017) is the direct precedent for treating disclosure volume as a process with its own institutional capacity constraints, separate from the underlying rate of software defects. Bostrom’s differential technological development framework (2014) is the background this paper’s broader project sits inside, though this paper does not extend into the normative territory Bostrom’s framework is built for.

IX. Provenance and What Remains Open

This revision replaces every housing statistic with numbers computed directly from FRED’s PERMIT, HOUST, and COMPUTSA series, pulled fresh for this draft, using ADF, KPSS, and AR(1) implementations written and run specifically for this project. The CVE 2018 figure is corrected against two sources independent of each other and of this project. The distillation exchange count

is corrected against Anthropic’s own disclosure, fetched directly. The AI capability doubling time is resolved in favor of the figure that traces to a specific saved table, over prose in an earlier draft.

What is not yet done: the cybersecurity series for 2019–2025 has not been re-verified point by point the way 2018 and the housing series were; the AI capability full-history (21-model) regression has not been re-run to confirm whether it should replace, or sit alongside, the 20-model figure used here; and the distillation efficiency parameters remain, honestly, unknown. A biotechnology-access domain drafted earlier in this project’s history has been removed from this paper entirely — it was never part of this paper’s scope, and belongs, if anywhere, in a separate project built around longevity governance instead of queueing dynamics.

No claim in this paper rests on a checksum, an audit badge, or a label saying a value was verified. Every number above either states the specific source it traces to, or states plainly that it has not yet been independently checked. That is the only form of evidence this project is now treating as evidence.